

Vivian (Jingcheng) Yu

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EDUCATION

UNIVERSITY OF WATERLOO

Waterloo, ON

Ph.D. in Actuarial Science

GPA: 91%

Sep 2024 - Present

- *Relevant Courses:* Theory of Probability 2, Deep Learning, Stochastic Differential Equation, Financial Econometrics
- **Teaching Certifications: Fundamentals of University Teaching Program; Certificate in University Teaching (In Progress)**

UNIVERSITY OF CALIFORNIA, LOS ANGELES

Los Angeles, CA

Master of Financial Engineering

GPA 3.82/4.0

Dec 2022

- *Relevant Courses:* Stochastic Calculus, Credit Risk, Econometric, Data Analysis & Machine Learning, Computational Methods in Finance, Derivatives, Fixed Income, Financial Risk Management
- *Activities:* Student Council Member

UNIVERSITY OF MANITOBA

Winnipeg, MB

B.S., Actuarial Mathematics/ Management

4.13/4.50 (Major GPA: 4.21/4.50)

Dec 2019

Honor: First Class Honor, Dean's Honor List

MEMBERSHIP & CERTIFICATIONS

- Fellow of the Society of Actuaries (FSA)
- Chartered Enterprise Risk Analyst of the Society of Actuaries (CERA)
- CFA Level II Candidate

RESEARCH & PROJECTS

Submodular Risk Measures (Preprint: arXiv:2603.01232)

Aug 2025 - Apr 2026

Advisor: Ruodu Wang

- Characterized submodularity for law-invariant convex risk measures: law-invariant coherent risk measures are submodular exactly when they are coherent distortion risk measures, including Expected Shortfall (ES).
- Gave a complete characterization for shortfall risk measures via the Arrow-Pratt measure of risk aversion, showed optimized certainty equivalents are always submodular, and proved adjusted ES is submodular only when it reduces to ES.
- Conducted an empirical illustration on daily US equity returns, finding no ES submodularity violations, many Value-at-Risk violations, and relatively few adjusted-ES violations.

ES-only E-backtesting for AR-GARCH Losses (Working Paper)

May 2026 - Present

Joint work with Ruodu Wang

- Developed sequential e-backtests for Expected Shortfall that require only reported ES forecasts, treating the associated VaR value as unobserved and taking infima over admissible values.
- Established e-process validity under the null hypothesis and constructed certified lower bounds for AR-GARCH losses via interval bounds with directed rounding.
- Designed predictable betting processes calibrated from finite-horizon crossing probabilities, and validated the procedures in simulations and an FX data analysis against VaR-ES benchmarks.

Mathematics of Scenario-based Risk Evaluation (Work in Progress)

Jun 2026 - Present

Joint work with Ruodu Wang and Chenxi Xia

- Studying risk measures determined by collections of scenarios (probability measures), covering scenario-based distortion and spectral risk measures.
- Developing structural and representation results for scenario-based risk evaluation, building on the framework of Wang and Ziegel (2021).

Classification and Decision Tree Model for Portfolio Design (UCLA Capstone Project in Financial Engineering)

March - Dec 2022

Academic Advisor: Lars Lochstoer (UCLA Anderson), Company Advisor: Gina Sanchez (Chantico Global)

- Collaborated in a multidisciplinary team guided by academic and industry advisors to analyze asset performance using CART models across 14 asset classes and 73 macroeconomic variables.
- Applied Rolling Window Analysis to enhance model robustness, consistently identifying 'Equity (Above Average)' and 'BAMLHE00EHYISYTW (D)' in S&P Global Infrastructure Total Return Index as assets with stable positive influence over different models and time periods.
- Presented findings that revealed the dynamic nature of feature importance in CART models, including variations in important features over different rolling periods and sensitivity to specific macroeconomic factors, offering valuable insights for adapting to different economic conditions.

Triggered Shorting Strategy on VIX (2022 IAQF Academic Competition Project)

May - Jun 2022

- Leadership in State Identification Models: Led a team to calibrate a 7-state Hidden Markov Model and a 3-state K-means clustering model, effectively dividing market performance into different states.
- Expertise in Neural Network Modeling: Employed Artificial Neural Networks using outputs from the cluster model, HMM model, and forward-stepwise-selection logit model, resulting in strong performance in predicting a binary market with only one error during the testing periods.
- High-Performing Short Strategy: Implemented a short strategy in identified negative market conditions, yielding an average return of 173.5%, which is 154% higher than the market return.

EXPERIENCE

UNIVERSITY OF WATERLOO

Waterloo, ON

Teaching Assistant

Sep 2024 - Present

- Facilitated Interactive Learning: Lead tutorials for undergraduate actuarial students, simplifying complex mathematical and risk management theories into accessible, engaging concepts.
- Student Mentorship: Provide personalized guidance during office hours, identifying individual learning gaps and adapting teaching strategies to support students with diverse learning styles.
- Curriculum Collaboration: Collaborate closely with professors to ensure the smooth delivery of course materials and fair assessment of student progress.

Academic Research Reviews and Grading (Society of Actuaries)

Remote

Reviewer

Aug 2022 – March 2024

- Served on the Review Committee for research paper evaluations by industry and academia.
- Contributed to various stages of the review process, ensuring high-quality assessments.
- Participated in the assessment of notable research papers such as 'Big Tech & Insurance' and 'Zero Knowledge Proof'.
- Acted as an end-of-module assessment grader for the Society of Actuaries, ensuring fair and thorough evaluations.

UNIVERSITY OF CALIFORNIA, LOS ANGELES

Los Angeles, CA

Teaching Assistant

Sep – Dec 2022

- Delivered tutorial lectures on stochastic calculus, ensuring comprehensive understanding.
- Provided academic support and guidance to students through grading assignments and proctoring exams.

ERNST & YOUNG GLOBAL CONSULTING SERVICES

New York, NY

Actuarial Consultant

Jan – Sep 2023

- Applied algorithmic optimization techniques in Python, including Simulated Annealing (SA) and Genetic Algorithm (GA), to optimize portfolios with liabilities and reinvestments.
- Developed a simulated algorithm efficiently solving complex problems while minimizing initial assets, which reduced financing cost by 50%.

CALIFORNIA DEPARTMENT OF INSURANCE

Los Angeles, CA

Actuarial Student Assistant

Jun – Sep 2022

- Conducted in-depth research on ISO filings and provided recommendations for revised guidelines.
- Consulted with ISO to address confusions and presented results to the department, improving on risk assessment.
- Redeveloped the Class Plan Application using VBA coding in Excel, resulting in a 30% reduction in processing time.

SKILLS

- AI & Efficiency: AI-Augmented Development (Gemini/Codex for rapid prototyping), LLM Integration
- Programming: Python (Proficient), R, SQL
- Libraries & Frameworks: pandas, NumPy, scikit-learn, TensorFlow, Tidyverse (R)
- Domain Knowledge: Financial Modeling, Risk Analysis, Stochastic Calculus